

STRING MILL

SPIRAL BLADE STRING MILL

MILLING

GENERAL DESCRIPTION

THE STRING MILL IS A SPIRAL-BLADED MILLING TOOL RUN IN THE STRING TO REMOVE DEBRIS, PERFORATION BURRS AND OTHER FOREIGN MATERIAL LEFT IN THE CASING ID AFTER DRILLING, WELLBORE CLEAN-UP AND PRE-COMPLETION RUNNING OPERATIONS. ITS ALIGNED FLUID CHANNELS GIVE MAXIMUM FLUID BY-PASS AND 360 DEGREE COVER FOR BETTER RECOVERY. IT IS PARTICULARLY SUITED TO PREPARING CASING WINDOWS BEFORE RUNNING LINERS OR COMPLETIONS. IT IS RUN ALONGSIDE A NON-ROTATING CASING SCRAPER AHEAD OF THE DRIFT SUB, REMOVING DRILLING FLUID RESIDUE, SCALE AND CEMENT WHILE RUNNING IN HOLE AND RECIPROCATING THE STRING. THE BLADES CAN BE FACED WITH A CHOICE OF HARDFACING, DEPENDING ON OPERATIONAL REQUIREMENTS.

TECHNICAL CHARACTERISTICS

Casing (in)	Weight (lb/ft)	Drift (in)	Max hard OD (in)	Body ID (in)	Overall length (in)	Neck length (in)	Neck OD (in)	Rig connection	Torsional yield (ft-lb)	Make-up torque (ft-lb)	Tensile (lb)
4-1/2	9.5 - 16.5	3.965 - 3.629	3.965 - 3.629	1.50	57	18	3.125	2-7/8 HT PAC	8,500	5,100	214,200
5-1/2	23 - 28.4	4.545 - 4.315	4.545 - 4.315	1.50	57	18	3.125	2-7/8 HT PAC	8,500	5,100	214,200
7	26	6.151	6.151	2.69	64	18	4.750	NC38	11,800	9,100	580,000
7	29	6.059	6.059	2.250	76	27.5	4.750	NC38	11,800	9,100	580,000
7	32	5.969	5.969	2.250	76	27.5	4.750	NC38	11,800	9,100	580,000
9-5/8	43.5	8.599	8.599	3.500	76.5	18	6.500	NC50	51,700	27,000	944,000
9-5/8	47	8.525	8.525	3.500	76.5	18	6.500	NC50	51,700	27,000	944,000
9-5/8	53.5	8.379	8.379	3.500	76.5	18	6.500	NC50	51,700	27,000	944,000
9-5/8	58.4	8.279	8.279	3.500	76.5	18	6.500	NC50	51,700	27,000	944,000
9-5/8	59.4	8.250	8.250	3.500	76.5	18	6.500	NC50	51,700	27,000	944,000

Other sizes, bespoke ODs and blade lengths on request. Dimensions in inches, loads in lb and ft-lb.

OPERATING LIMITS

Casing (in)	Max run speed (ft/min)	Max rotating speed (rpm)	Max compression when rotating (lb)	Max pressure test (psi)
4-1/2, 5-1/2	60	30	5,000	10,000
7	60	120	30,000	10,000
9-5/8	60	120	40,000	10,000

BENEFITS

- **High fluid by-pass:** fluid and debris flow past the tool body.
- **Compatible:** large inner bore, works with all standard clean-up tools and prevents surge and swab.
- **Robust:** no internal or external screws or bolts.
- **Flexible:** API and non-standard casing sizes, bespoke ODs and blade lengths.

HOW TO ORDER

Please specify: 1. casing size and weight 2. rig connection 3. blade length and hardfacing 4. other tools run in the string (scraper, drift sub).